

# Blood Sugar Battle

## Teacher Key + Manual Scoring Guide

Concept-aligned version: 12 questions, 30 possible points. Students are not required to recall pathway names.

### Blood Glucose Control

|                       |                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 1 (2 pts) - A student drinks a large sweetened coffee on an empty stomach. What happens to blood glucose, and which hormone responds? |
| <b>Correct answer</b> | Blood glucose rises. Insulin is released and helps move glucose into cells for energy or storage, bringing blood glucose back down.   |
| <b>Scoring</b>        | 1 pt: blood glucose rises; 1 pt: insulin responds and lowers blood glucose through uptake/storage.                                    |

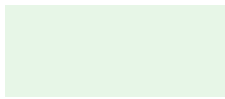
|                       |                                                                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 2 (2 pts) - A person has not eaten for several hours and blood glucose begins to fall. Which hormone responds, and what stored carbohydrate can help maintain blood glucose? |
| <b>Correct answer</b> | Glucagon responds. The liver can break down its stored glycogen and release glucose into the bloodstream.                                                                    |
| <b>Scoring</b>        | 1 pt: glucagon; 1 pt: liver glycogen is used to help raise/maintain blood glucose.                                                                                           |

|                       |                                                                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 3 (3 pts) - After a large carbohydrate meal, immediate energy needs are met. Where is extra glucose stored first, and what can happen after glycogen storage is filled?  |
| <b>Correct answer</b> | Extra glucose is stored as glycogen in the liver and muscles first. When glycogen storage is filled and energy is still excessive, the excess can be stored as body fat. |
| <b>Scoring</b>        | 1 pt: glycogen first; 1 pt: liver and/or muscle storage; 1 pt: additional excess can be stored as fat.                                                                   |

### Storage & Fasting

|                       |                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 4 (2 pts) - Why can liver glycogen help maintain blood glucose between meals, while muscle glycogen cannot?                                          |
| <b>Correct answer</b> | Liver glycogen can be broken down and glucose released into the blood for other tissues. Muscle glycogen is reserved for that muscle's own activity. |
| <b>Scoring</b>        | 1 pt: liver can support blood glucose; 1 pt: muscle glycogen is used locally by muscle.                                                              |



|                       |                                                                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 5 (2 pts) - A person starts a very-low-carbohydrate diet and loses several pounds during the first week. What explains much of the early weight loss? |
| <b>Correct answer</b> | Much of the early loss is water. Glycogen is stored with water, so when glycogen stores are depleted, water is lost too.                              |
| <b>Scoring</b>        | 1 pt: water loss; 1 pt: connects water loss to depletion of glycogen stores.                                                                          |

## Food Form & Glycemic Response

|                       |                                                                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 6 (2 pts) - Which would usually cause a slower rise in blood glucose: whole-grain bread or refined white bread? Explain why.              |
| <b>Correct answer</b> | Whole-grain bread usually causes a slower rise because it contains more fiber, which slows carbohydrate digestion and glucose absorption. |
| <b>Scoring</b>        | 1 pt: whole-grain bread; 1 pt: fiber slows digestion/absorption or reduces the glucose spike.                                             |

|                       |                                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 7 (2 pts) - Which would usually raise blood glucose more quickly: a whole apple or apple juice? Explain.                                     |
| <b>Correct answer</b> | Apple juice usually raises blood glucose more quickly because it contains little fiber. The fiber in a whole apple slows glucose absorption. |
| <b>Scoring</b>        | 1 pt: apple juice; 1 pt: whole fruit fiber slows absorption.                                                                                 |

|                       |                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 8 (2 pts) - Compare a plain bagel with a bagel eaten with peanut butter. Which would generally produce the slower blood-glucose rise, and why? |
| <b>Correct answer</b> | The bagel with peanut butter. Protein and fat slow digestion and glucose absorption, producing a more gradual blood-glucose response.          |
| <b>Scoring</b>        | 1 pt: bagel with peanut butter; 1 pt: protein/fat slow digestion or glucose absorption.                                                        |

|                 |                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b> | 9 (2 pts) - Honey is stirred into oatmeal, maple syrup is poured on pancakes, and agave is added to tea. Which of these count as added sugars, and why? |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

|                       |                                                                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Correct answer</b> | All three count as added sugars because they were introduced during preparation or at the table. Their natural origin does not make them naturally occurring sugar in that food. |
| <b>Scoring</b>        | 1 pt: all three; 1 pt: recognizes that sugar added during preparation/at the table is added sugar regardless of source.                                                          |

## Diabetes & Fiber

|                       |                                                                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 10 (3 pts) - A person makes insulin, but the body's cells do not respond to it effectively. What happens to blood glucose, and which type of diabetes is this most associated with? |
| <b>Correct answer</b> | Glucose does not enter cells normally, so blood glucose stays elevated. This insulin resistance is most associated with Type 2 diabetes.                                            |
| <b>Scoring</b>        | 1 pt: cells respond poorly to insulin; 1 pt: blood glucose remains elevated; 1 pt: Type 2 diabetes.                                                                                 |

|                       |                                                                                                                                                                                                                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 11 (3 pts) - Give two ways a high-fiber food can support health, and name one food that is a good source of fiber.                                                                                                                                                                     |
| <b>Correct answer</b> | Acceptable effects include slowing glucose absorption, helping bowel regularity by holding water/adding stool bulk, helping lower cholesterol, increasing fullness, or feeding beneficial gut microbes. Good sources include beans, whole grains, vegetables, fruits, nuts, and seeds. |
| <b>Scoring</b>        | 1 pt each for two correct fiber benefits; 1 pt for one appropriate high-fiber food source.                                                                                                                                                                                             |

## Final Boss

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>       | 12 (5 pts) - Jordan eats pancakes with syrup and orange juice at 8:00 a.m., skips lunch, and does a hard workout late in the afternoon. Describe what happens to blood glucose and fuel storage/use from breakfast through the workout.                                                                                                                                                                                                                                                                      |
| <b>Correct answer</b> | After breakfast, blood glucose rises and insulin increases. Glucose is used for immediate energy and stored as glycogen; excess energy can be stored as fat. As time passes without food, blood glucose begins to fall, glucagon increases, and the liver uses glycogen to help maintain blood glucose. During the workout, muscle glycogen is used by the working muscles. With prolonged low carbohydrate availability, the body relies more on fat and other alternative fuels, and ketones may increase. |
| <b>Scoring</b>        | Up to 5 pts: blood glucose/insulin rise after breakfast; glucose used/stored as glycogen; glucagon/liver glycogen between meals; muscle glycogen fuels exercise locally; prolonged fasting shifts toward fat/alternative fuels or ketones.                                                                                                                                                                                                                                                                   |

## Tie-Breaker + Running the Game

**Tie-breaker prompt:** A student eats dinner at 7:00 p.m., skips breakfast, and does a hard workout at 7:00 a.m. Explain which stored fuel helps maintain blood glucose overnight and which stored fuel primarily supports the working muscles during the workout.

**Tie-breaker answer:** Liver glycogen helps maintain blood glucose between meals. Muscle glycogen is used locally by the working muscles. As fasting continues and carbohydrate availability stays low, the body increasingly relies on fat and other alternative fuels.

- Suggested timing: 15-20 minutes total. Use about 30-45 seconds for most questions, 60 seconds for 3-point questions, and up to 2 minutes for the Final Boss.
- Create groups of 3-5 students and give each group one answer sheet and one pen/pencil.
- Project the game at the front of the room. Keep this teacher key with you.
- Show one question at a time. Groups discuss and write one final response before you reveal anything.
- Click Reveal Answer. Read the scoring line aloud. Groups can self-score, or swap sheets with a neighboring group for scoring.
- For selected questions, ask one group to explain why before moving on. If many groups miss the same idea, do a 30-second reteach.
- After Question 12, total the score out of 30. Use the tie-breaker only if needed.
- Scoring philosophy: Accept equivalent wording that demonstrates the concept. Do not require students to name glycogenolysis, gluconeogenesis, glycogenesis, lipogenesis, or ketogenesis. You may introduce those terms during the debrief after students explain what the body is doing.